

DATA ANALYSIS & EMPIRICAL METHODS

Detailed Presentation & Speech Guide for 10 Key Problems

This guide is a comprehensive presentation script designed to be read or referenced while presenting the solutions to the 10 statistics and data analysis problems in the 07_data_analysis directory. It highlights core concepts, empirical findings, methodologies, and critical takeaways for each problem.

Each section includes:

1. Problem Objective & Scope: Brief summary of what the problem aims to achieve.
2. Key Data & Statistics: Numeric findings generated from the data files.
3. Speech Transcript: Word-for-word detailed speech script for the presenter.
4. Presenter Tip: Strategic advice for slide visualizations and audience engagement.

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SLIDE 1: Problem 1: Die Rolls and Empirical Distribution

1. PROBLEM OBJECTIVE & SCOPE

Compute empirical frequencies of 1000 simulated die rolls, compare them with theoretical fair die probabilities ($1/6$), and evaluate whether the die appears to be fair based on the data.

2. KEY DATA & STATISTICS

- Sample size: 1000 rolls (Seed: 701)
- Absolute Frequencies: 1->157, 2->174, 3->151, 4->184, 5->158, 6->176
- Empirical Probabilities: $P(\text{even}) = 0.534$, $P(\geq 5) = 0.334$, $P(=6) = 0.176$
- True generating probabilities (biased die): [0.14, 0.16, 0.17, 0.18, 0.16, 0.19]

3. SPEECH TRANSCRIPT

Ladies and gentlemen, in our first slide, we discuss 'Die Rolls and Empirical Distribution'. We analyze a dataset containing 1000 simulated die rolls. First, let's look at the data structure: each row represents a single die roll trial. Counting the absolute frequencies, we find that face 4 appeared most frequently (184 times) while face 3 appeared least frequently (151 times). By dividing these counts by the total 1000 rolls, we obtain empirical probabilities.

For instance, the empirical probability of getting an even number is 0.534, at least a 5 is 0.334, and exactly a 6 is 0.176. Theoretically, for a fair die, these would be 0.500, 0.333, and 0.167 respectively. These deviations are natural results of random sampling variation. We conducted a formal Chi-Squared goodness-of-fit test and obtained a statistic of 5.128. Since this is well below the critical value of 11.07 (at a 5% significance level), an external observer cannot reject the fairness hypothesis based solely on this sample. However, we know the die is actually biased because we set its true probabilities to [0.14, 0.16, 0.17, 0.18, 0.16, 0.19]. This highlights that detecting subtle biases often requires much larger sample sizes.

PRESENTER TIP: Point out the red dashed line on the bar chart representing the fair-die probability of 0.167 and mention the Chi-Squared statistic (5.128) to show an analytical approach.

SLIDE 2: Problem 2: Coin Tosses and Relative Frequencies

1. PROBLEM OBJECTIVE & SCOPE

Track the behavior of relative frequencies over 2000 coin tosses and demonstrate the Law of Large Numbers (LLN) in practice.

2. KEY DATA & STATISTICS

- Sample size: 2000 tosses (Seed: 702)
- Results: Heads (H) = 1038 (51.9%), Tails (T) = 962 (48.1%)
- True generating probabilities: Heads = 0.52, Tails = 0.48

3. SPEECH TRANSCRIPT

In our second slide, we examine one of the cornerstones of probability theory: the Law of Large Numbers, through 2000 coin tosses. Our trial results yield 1038 Heads (51.9%) and 962 Tails (48.1%).

The key visual here is the line plot showing the relative frequency of Heads against the number of trials. In the first few tosses, the frequency fluctuates wildly (e.g., getting 80% Heads in the first 10 trials is very common). As the number of trials increases, these fluctuations diminish and the frequency converges to a stable value. This stabilization is the Law of Large Numbers in action. Our final frequency of 51.9% is close to a fair 50%, but actually, the coin was generated with a true probability of 52.0%. The frequency converges to the coin's true parameter, not the idealized 50%. It is vital to explain to the audience that empirical relative frequency is an observed statistic, whereas theoretical probability is an intrinsic parameter of the system.

PRESENTER TIP: Point to the stabilization trend on the chart to illustrate LLN, explaining how random noise averages out over time.

SLIDE 3: Problem 3: Exam Scores in Two Groups

1. PROBLEM OBJECTIVE & SCOPE

Compare student exam scores in groups A and B using measures of central tendency and dispersion, illustrating the limits of using averages alone.

2. KEY DATA & STATISTICS

- Overall Mean / Median: 68.22 / 69.00
- Group A (Homogeneous): Mean = 72.15, Median = 73.00, Std Dev = 11.45, Pass Rate = 85%
- Group B (Heterogeneous): Mean = 64.29, Median = 65.00, Std Dev = 15.12, Pass Rate = 71%

3. SPEECH TRANSCRIPT

Our third slide compares student exam scores across two groups, illustrating why looking only at averages can be misleading. Group A has a mean score of 72.15, and Group B has 64.29. Group A seems better.

However, checking dispersion, Group A's standard deviation is 11.45, while Group B's is 15.12. This means Group B's scores are much more spread out (highly heterogeneous). B has both high achievers and struggling students, whereas Group A is more consistent (homogeneous). Due to B's high variance, its pass rate is only 71% compared to Group A's 85%. The takeaway is that decision-makers must look at variability (variance/std dev), not just averages, to avoid missing struggling or high-potential students.

PRESENTER TIP: Highlight the wider Interquartile Range (IQR) and whiskers in Group B's boxplot to visually support the variance argument.

SLIDE 4: Problem 4: Delivery Times, Skewness, and Outliers

1. PROBLEM OBJECTIVE & SCOPE

Analyze a right-skewed distribution of delivery times, explain the difference between mean and median, and identify outliers using the $1.5 \times \text{IQR}$ rule.

2. KEY DATA & STATISTICS

- Sample size: 120 deliveries
- Mean / Median: 38.45 min / 32.10 min (Difference: 6.35 min)
- Q1 / Q3 / IQR: 25.50 min / 45.80 min / 20.30 min
- Outlier threshold: $Q3 + 1.5 \times \text{IQR} = 76.25$ min (8 outliers found)
- Delayed rate (>50 min): 17.5%

3. SPEECH TRANSCRIPT

Our fourth slide covers delivery times and outliers. The mean delivery time is 38.45 minutes, while the median is 32.10 minutes. The fact that the mean is significantly larger than the median indicates a right-skewed distribution. Most deliveries are completed within normal limits, but occasional extreme delays (due to traffic, weather, or accidents) pull the mean upward, while the median remains robust.

To identify outliers, we apply the $1.5 \times \text{IQR}$ rule. Adding 1.5 times the IQR (20.30) to Q3 (45.80) yields an upper threshold of 76.25 minutes. Any delivery exceeding this is an outlier; we found 8. Additionally, 17.5% of deliveries were delayed. For skewed distributions, the median is a more robust indicator of central tendency than the mean.

PRESENTER TIP: Reference the long right tail of the histogram and explain the concept of robustness using a simple analogy (like how adding a billionaire to a room increases the average income but leaves the median unchanged).

SLIDE 5: Problem 5: Customer Survey and Conditional Frequencies

1. PROBLEM OBJECTIVE & SCOPE

Investigate the relationship between customer satisfaction and renewal rates using conditional probabilities.

2. KEY DATA & STATISTICS

- Sample size: 500 customers
- Overall Renewal Rate: 64.2%
- Renewal by Channel: Email (72%), Phone (58%), Web (66%)
- Conditional Probabilities: $P(\text{renewed} \mid \text{satisfaction}=5) = 88.5\%$, $P(\text{satisfaction}=5 \mid \text{renewed}) = 35.6\%$

3. SPEECH TRANSCRIPT

On slide five, we merge marketing analytics and conditional probability. Among 500 surveyed customers, the overall renewal rate is 64.2%. Email is the most effective channel with a 72% renewal rate, while Phone is the lowest at 58%.

The core statistical point is the distinction between two conditional probabilities: $P(\text{renewed} \mid \text{satisfaction}=5)$ which is 88.5%, and $P(\text{satisfaction}=5 \mid \text{renewed})$ which is 35.6%. The first tells us that 88.5% of highly satisfied customers renew, showing satisfaction's predictive power. The second tells us that only 35.6% of those who renewed had a satisfaction score of 5, indicating that renewed customers also include satisfied (4) or neutral (3) customers. The relationship between these two probabilities is defined by Bayes' Theorem.

PRESENTER TIP: Write down the two conditional probability terms side by side to emphasize how reversing the condition changes the question being answered.

SLIDE 6: Problem 6: Factory Measurements and Specification Limits

1. PROBLEM OBJECTIVE & SCOPE

Evaluate length measurements from two machines against specification limits (98mm - 102mm) to highlight the role of variance in quality control.

2. KEY DATA & STATISTICS

- Target / Limits: 100.00 mm / 98.00 - 102.00 mm
- Overall Spec Compliance: 92.4%
- Machine 1 (Stable): Mean = 100.02 mm, Std Dev = 0.45 mm, Compliance = 99.2%
- Machine 2 (Unstable): Mean = 99.85 mm, Std Dev = 1.20 mm, Compliance = 82.5%

3. SPEECH TRANSCRIPT

Slide six focuses on industrial quality control. The target length of produced parts is 100 mm, with acceptable limits between 98 and 102 mm. Overall compliance is 92.4%, but individual machine analysis reveals a key lesson.

Machine 1 has a mean of 100.02 mm and a very small standard deviation of 0.45 mm, yielding a 99.2% compliance rate. Machine 2 has a mean of 99.85 mm, which is close to the target, but its standard deviation is 1.20 mm! This high variability causes 17.5% of its parts to fall outside limits, resulting in significant scrap. In manufacturing, keeping variability low (small standard deviation) is the true key to quality, not just keeping the mean close to target.

PRESENTER TIP: Point to the boxplot of Machine 2 to show its wide spread, and mention that standard deviation directly corresponds to defect rate and manufacturing waste.

SLIDE 7: Problem 7: Call Center Requests Over Time

1. PROBLEM OBJECTIVE & SCOPE

Analyze hourly call volumes, compare empirical mean and variance to assess Poisson suitability, and examine variations over time.

2. KEY DATA & STATISTICS

- Hourly Mean / Variance: 14.2 / 15.1 (Highly similar)
- Weekday / Weekend Mean: 16.5 / 8.4
- Peak hours: 10:00 - 14:00

3. SPEECH TRANSCRIPT

In slide seven, we model call center traffic. The hourly mean is 14.2, and the variance is 15.1. In a Poisson distribution, the mean and variance are equal. Their closeness here suggests that a Poisson model is an excellent candidate.

However, we cannot use a single Poisson process for the entire dataset because call arrival rates vary significantly. Weekday traffic (mean 16.5) is double the weekend traffic (mean 8.4), and hourly traffic peaks between 10:00 and 14:00. We must model this as a Non-homogeneous Poisson Process where the rate parameter varies by hour and day type to optimize staffing.

PRESENTER TIP: Explain that using a single average (14.2) for staffing would cause severe understaffing during peak hours (10:00-14:00) and waste resources at night.

SLIDE 8: Problem 8: Waiting Times and Empirical CDF

1. PROBLEM OBJECTIVE & SCOPE

Compare waiting times for standard vs priority services and use the Empirical CDF (ECDF) to estimate waiting probabilities.

2. KEY DATA & STATISTICS

- Overall Mean / Median: 11.2 min / 8.5 min
- Priority Service: Mean = 4.1 min, Median = 3.2 min
- Standard Service: Mean = 13.5 min, Median = 11.2 min
- ECDF Probabilities: $P(T \leq 5 \text{ min}) = 0.45$, $P(T \leq 10 \text{ min}) = 0.68$, $P(T > 20 \text{ min}) = 0.12$

3. SPEECH TRANSCRIPT

Slide eight introduces the Empirical Cumulative Distribution Function, or ECDF, using customer waiting times. Waiting times are typically right-skewed, similar to an exponential distribution. Priority service waiting times (mean 4.1 minutes) are much lower than standard service (mean 13.5 minutes).

The ECDF allows us to read probabilities directly from the data without parametric assumptions. We can estimate that 45% of customers wait 5 minutes or less, 68% wait 10 minutes or less, and only 12% experience extreme waits of more than 20 minutes. This is useful for service level agreements (SLAs).

PRESENTER TIP: Point out that the y-axis of the ECDF represents the cumulative probability, ranging from 0 to 1, showing the fraction of customers served within the x-axis time.

SLIDE 9: Problem 9: Warehouse Orders, Demand, and Returns

1. PROBLEM OBJECTIVE & SCOPE

Analyze order quantities and return rates across 3 warehouses and 5 product groups to identify operational risks.

2. KEY DATA & STATISTICS

- Total Order Quantity: 45,210 units
- High Demand: Electronics (18,200), Apparel (12,400)
- Overall Return Rate: 14.8%
- Returns by Group: Apparel (22.1%), Electronics (12.4%)
- Returns by Warehouse: Northern (16.2%), Southern (13.1%)

3. SPEECH TRANSCRIPT

Slide nine covers demand and return rates. While Electronics has the highest demand (18,200 units), the return rate is our key metric for profitability. The overall return rate is 14.8%, but Apparel has a massive return rate of 22.1%. This represents a significant shipping and handling cost.

Additionally, the Northern Warehouse has the highest return rate at 16.2%. This suggests we need to review apparel size charts and Northern Warehouse packaging/shipping operations to reduce return rates and improve efficiency.

PRESENTER TIP: Contrast the high sales volume of Apparel with its high return rate, explaining how returns erode margins.

SLIDE 10: Problem 10: Correlation Traps

1. PROBLEM OBJECTIVE & SCOPE

Demonstrate how correlation (r) can be distorted by outlier groups and emphasize the necessity of scatter plots.

2. KEY DATA & STATISTICS

- Full Dataset Correlation (r): 0.82 (Strong positive correlation)
- Without Outliers Correlation (r): -0.15 (Weak negative correlation)
- Finding: A single outlier group in the top right distorts the r calculation, hiding the actual negative trend of the main cluster.

3. SPEECH TRANSCRIPT

Our final slide illustrates correlation traps. The overall Pearson correlation coefficient between X and Y is $r = 0.82$, suggesting a strong positive relationship. However, looking at the scatter plot, we see that 95% of the data points actually form a cloud with a weak negative trend.

A small outlier group in the top-right corner pulls the mathematical calculation, creating the illusion of a positive relationship. Removing this group drops the correlation to $r = -0.15$. This demonstrates that relying on summary statistics alone without visual inspection can lead to incorrect conclusions.

PRESENTER TIP: Point to the scatter plot to show the outlier group vs the main cluster, referencing Anscombe's Quartet to conclude the presentation on a strong note.